

Evaluating AI-based visual privacy through Cartoonification: Impact on expressive participation and consent for data retention

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ARTICLE INFO

Keywords:

Visual privacy
Privacy-preserving educational technology
Learning analytics
Emotion detection
Artificial intelligence in education
Sensitive topic discussions
Ethical AI in education

ABSTRACT

Learning analytics tools embedded with video devices are increasingly prevalent for capturing student interactions during learning activities. This approach is useful for classroom observation and analyzing students' emotional expression and perceived confidence during discussions. However, students expressed concern about how their identities could be represented, interpreted, and misused during these analytic and research processes. These privacy concerns are heightened during video-enabled classroom discussions of sensitive topics, as students may feel vulnerable about being recorded for analysis. This study explored the impact of visual privacy through video cartoonification on students' expressive participation during class discussions on sensitive topics, and their willingness to have their video recorded and used for analysis. Using a multi-method approach, we conducted a study in which students participated in discussions on topics such as police brutality, AI and racial bias, and science-religion tensions, using a platform with an optional cartoonification feature. Data collection included surveys assessing participants' comfort and confidence levels and AI-powered emotional tone analysis of discussion transcripts. The results revealed a strong positive correlation between visual privacy and students' socio-emotional participation levels, reflected in their emotional tone when expressing opinions. Notably, there was a statistically significant difference ($p = 0.001$) in the expression of concern-emotional tone across all topics when cartoonification was active. Also, this study demonstrates that visual privacy through cartoonification can effectively address students' concerns about visibility in video-enabled learning analytics while maintaining the benefits of capturing student interactions for analysis. The findings provide practical solutions for balancing learning analytics with student privacy concerns.

1. Introduction

The integration of multimodal learning analytics (MMLA) systems into the learning environment has been instrumental in enhancing the collection and analysis of student data for understanding learning patterns, engagement, and interactions. This provides the meaningful insights required to support students' learning activities and improve learning outcomes. Video-based MMLA systems have emerged as powerful tools for capturing rich, multimodal data that reveal patterns of engagement, communication, and interaction, useful for evaluating students' learning. Research has demonstrated that video data collection during lectures, group activities, and presentations provides valuable data for educational research and learning analytics [1–3]. These data have proven valuable in classroom observation studies, where video

data enabled researchers to explore the correlations between behavioral and emotional aspects and cognitive engagement [4].

Classroom or group discussion activities are foundational to active learning because they create dynamic environments in which student engagement, critical thinking, and collaborative knowledge construction naturally emerge [5–8]. Through peer interaction, students articulate perspectives, negotiate meaning, and deepen their understanding of the subject. These dialogic exchanges, whether recorded or not, make reasoning processes, idea-building, and engagement visible within the learning community. As such, the pedagogical value of discussion lies in its capacity to foster reflective thinking, social learning, and the construction of shared knowledge independent of analytic instrumentation.

From a learning analytics perspective, recording these discussions enables systematic examination of interactional processes that extend

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<https://doi.org/10.1016/j.caeo.2026.100393>

Received 22 April 2025; Received in revised form 28 February 2026; Accepted 12 July 2026

Available online 13 July 2026

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beyond real-time observation. Video-based multimodal learning analytics (MMLA) systems capture both verbal and nonverbal behaviors, allowing researchers to analyze participation patterns, emotional responses, and group dynamics with greater granularity [9,10]. Such recordings provide multimodal data that support investigations into correlations between behavioral, emotional, and cognitive engagement in educational settings [11,12,4]. In this sense, the analytic value of recording is distinct from the pedagogical value of discussion; recording serves as a research and evaluation mechanism to understand collaborative learning processes.

While both discussion pedagogy and analytic recording offer important benefits, challenges arise when classroom discussions focus on sensitive or controversial topics. Even in unrecorded settings, some students may hesitate to speak openly due to the potential social, cultural, or personal implications of their views [13]. Such topic-related reluctance can affect participation dynamics and the depth of dialogue, despite the pedagogical importance of engaging with complex or contested issues. Discussions on sensitive topics are critical for fostering critical thinking and meaningful dialogue about academic, social, and civic issues [14–16]. Yet students may respond with uncomfortable silences or more cautious contributions, particularly when the social stakes of the topic are perceived as high.

Beyond the inherent sensitivities of certain topics, additional hesitation may arise in video-enabled environments [17,18]. Research has shown that students often provide more superficial responses or become more silent when discussions are recorded [19]. This reluctance may stem from uncertainty about identity exposure, as students become conscious of attaching their facial expressions and voices to potentially controversial viewpoints [20,21,10]. Additionally, prior research indicates that students are particularly sensitive to the long-term implications of their video data being retained for posthoc analysis, shared among researchers, or repurposed for broader educational studies [22–24]. These concerns include the risk of loss of anonymity, breach of confidentiality, and potential misuse for commercial or unintended purposes [25,26]. Moreover, another primary concern students expressed is not merely being recorded but rather how their video data are stored, distributed, and used after class activities [25]. Consequently, while recording enhances analytical visibility, it may also alter participation conditions, posing a distinct challenge for video-based learning analytics systems.

While video-based learning analytics offer valuable insights into collaborative engagement, their effectiveness depends on learners' willingness to participate under conditions of perceived privacy. Privacy concerns may discourage students from engaging fully in video-recorded discussions, particularly when topics are sensitive, thereby constraining the expressive behaviors that analytics systems seek to understand. This tension creates a paradox: the analytic value of video data increases with richer participation, yet the visibility inherent in recording may inhibit the very dialogue it aims to capture.

Given these challenges, it becomes imperative to explore interventions that enhance students' expressive participation in sensitive-topic discussions, provide a sense of privacy, and balance the need to collect valuable video data on student engagement. We integrated a visual privacy-enhancing feature, specifically a cartoonification-based visual transformation [27], into an existing learning analytics system. This enabled the transformation of students' faces into stylized cartoon-like representations that differ substantially from the original appearance while preserving expressive cues. While other privacy-enhancing techniques, such as blurring or removing videos altogether, can obscure identity [28], they often diminish the multimodal insights on which video-based learning analytics rely. By mitigating concerns about identity exposure, cartoonification may encourage more open, expressive participation during recorded discussions, thereby supporting ethical implementations of video-based learning analytics.

This study investigates how visual privacy through video

cartoonification can address two critical concerns in learning analytics: students' willingness to be recorded and their socio-emotional expressiveness during challenging classroom dialogues, while maintaining the benefits of video-based analysis. Therefore, this study seeks to answer the following research questions:

1. How does visual privacy through cartoonification influence students' willingness to have their data captured and collected for further analysis and educational research purposes?
2. What impact does video cartoonification have on student expressive participation and confidence to express opinions during sensitive topic discussions?

To explore this research question, we employed a multi-method approach. We conducted an experimental study to examine whether video cartoonification can address privacy concerns, enhance expressive participation, and influence students' willingness to allow their video data to be stored. The study involved engaging students in discussions on sensitive and controversial topics, including police brutality and racism, AI and racial bias, and tensions between science and religion. These topics were deliberately chosen because they tend to elicit student hesitation in recorded settings, making them an ideal context for examining the impact of cartoonification. Additionally, students were informed that video data collected during discussions of sensitive topics would be stored and used for further analysis. We used the Building Literacy In N-person Collaboration (BLINC) platform [29], which allows the cartoonification feature to be activated or deactivated during video-based discussions. The participants engaged in discussions with the feature on and off, enabling a comparative analysis. Following each discussion session, the participants completed a comprehensive survey. This questionnaire elicited subjective feedback on their comfort and confidence levels when using the platform to discuss sensitive topics, and on their willingness to have their video data stored and used for posthoc analysis.

Therefore, the unique contribution of this research is the introduction of a privacy-aware participation design approach for video-based learning analytics by integrating an expression-preserving cartoonification transformation into a synchronous discussion environment. Rather than treating privacy as a trade-off that suppresses analytic value, the transformation retains salient expressive facial cues, such as affective dynamics and interactional signals, while reducing identifiable visual detail. This design contributes to more ethically grounded use of video data by allowing learning analytics systems to operate on privacy-transformed representations that are structurally compatible with learning analytics interpretation. Building on prior stylized visual transformation models that retain important facial cues [30], the present work operationalizes this capability in a learning context and examines its human-centered implications. Specifically, we examine how a privacy-aware visual layer influences students' expressive participation, self-reported confidence in sharing opinions during sensitive discussions, and willingness to allow future data use in learning analytics contexts. The transformation is intended to mitigate perceived visibility risks in routine educational settings rather than to serve as a formally validated defense against human or automated adversarial re-identification. Therefore, the primary privacy concern addressed is students' perceived visibility and identifiability by instructors, peers, and researchers during or after recorded discussions, particularly in sensitive-topic contexts. Through this approach, our study provides empirical evidence that visual privacy interventions can mitigate privacy concerns and support expressive socio-emotional participation during sensitive collaborative discussions.

2. Literature review

The intersection of learning analytics, video technology, and privacy concerns in educational settings has garnered significant attention in

recent years. This literature review examines the current state of research on the use of video-based learning analytics tools and privacy preservation methods in educational contexts. Moreover, it explored the ethical and privacy issues arising from the storage, use, and distribution of student video data in educational research, classroom observation, and learning analytics. While substantial work has been done in these areas, there remains a critical gap in understanding how privacy-enhancing techniques, particularly visually transformative approaches such as cartoonification, influence students' willingness to participate in learning activities where their video data are collected for educational research and learning analytics. The visual privacy feature affects students' expressive participation in discussions of controversial topics. These investigations are grounded in various theoretical frameworks, including privacy regulation and social constructivism. This review highlights several key themes: 1) privacy concerns in the introduction of invasive educational technologies to learning analytics, 2) the growing use of video-based learning analytics tools and associated privacy concerns, 3) the impact of visual privacy on student engagement, and 4) the need for research to investigate how visual privacy-enhancing features impact both students' willingness to participate in video data collection and their confidence and emotional expressiveness during challenging classroom discussions. By synthesizing these themes, this review sets the stage for the current study, which aims to evaluate the influence of cartoonification on students' willingness to have their data collected and their confidence in freely expressing their opinions during sensitive discussions.

2.1. Theoretical framework

This study is grounded in two complementary theoretical perspectives: Privacy Regulation Theory (PRT) [31,32] and Social Constructivism [33]. While Privacy Regulation Theory explains how individuals manage self-disclosure under conditions of perceived surveillance or identity exposure, Social Constructivism explains how knowledge construction depends on active, expressive participation in social interaction. Taken together, these frameworks allow us to examine how perceived identity exposure in video-mediated contexts may shape expressive participation in collaborative learning environments.

1. **Privacy Regulation Theory:** This theory defines privacy as a balance between different states (e.g., anonymity, reserve) and functions (e.g., personal autonomy, emotional release). Individuals regulate their self-disclosure based on perceived privacy risks, especially in situations involving surveillance or exposure. Video recording of learning activities, particularly in controversial classroom discussions, may create circumstances in which students seek anonymity to avoid identification and public scrutiny. In particular, the captured data may be used outside the classroom context for educational research, classroom observation, or video learning analytics. Within this framework, students' reluctance to consent to video data collection reflects concern for personal autonomy, as they seek to protect their identities from exposure or misuse [31]. Cartoonification functions in this study as a visual boundary-regulation mechanism, potentially shifting students toward a more comfortable disclosure state by reducing identifiable detail while maintaining interactive engagement in discussions. By applying privacy regulation theory, this study investigates whether cartoonification can help students feel more in control of their privacy, thereby encouraging candid emotional expression and confidence in sensitive discussions without fear of being identified. Therefore, in this research, perceived privacy control was examined through participants' self-reported comfort, confidence, and willingness to allow video data collection, providing measurable indicators of privacy regulation processes.
2. **Social Constructivism:** This theory posits that learning is a social process in which knowledge is constructed through peer interaction

and discussion. Classroom discussions are a critical component of active learning, requiring students to exchange ideas, challenge perspectives, and collaboratively build knowledge. However, privacy concerns can disrupt this learning process by inhibiting student engagement and expression, especially in video-recorded discussions of sensitive topics. If students are concerned about being recorded, they may choose to remain silent or provide superficial responses, thereby reducing the depth of discussion and limiting collaborative knowledge construction [6,8]. This study extends social constructivist theory by examining how cartoonification can foster a psychologically safe environment in which students feel comfortable expressing their views without fear of exposure. By allowing students to maintain visual privacy while engaging in discussion, cartoonification supports the principles of socio-emotional participation and ensures that privacy concerns do not hinder students' critical thinking. Consequently, this study operationalized expressive participation through the frequency of emotional tone in discussion transcripts, reflecting how learners externalized thoughts and feelings during collaborative dialogue.

Importantly, Privacy Regulation Theory and Social Constructivism are not applied independently but are theoretically integrated in this study. Privacy Regulation Theory explains the mechanism (perceived identity exposure influencing disclosure behavior), while Social Constructivism explains the learning consequence (how disclosure behavior shapes expressive participation). Thus, this study focuses on socio-emotional indicators of participation, including confidence in expressing opinions and emotional expressiveness during discussions of sensitive topics. By linking privacy boundary management to expressive participation in dialogue, we examine how visual privacy may recalibrate participation dynamics in sensitive-topic discussions.

Based on these mappings, we expect that cartoonification would be associated with greater perceived privacy control and more expressive participation in sensitive-topic discussions. These operationalized constructs provide a conceptual lens for examining how privacy regulation mechanisms may shape expressive participation within socially constructed learning interactions. Specifically, drawing on Privacy Regulation Theory, we expect that participants in the cartoon-on condition would report higher levels of perceived privacy control, reflected in greater comfort, confidence, and willingness to allow video recording and storage. Drawing on Social Constructivism, we further expected that reduced perceived identity exposure would be associated with increased expressive participation, operationalized through higher frequencies of emotional tone in discussion transcripts.

2.2. Privacy concerns in the use of invasive educational technology

Traditional student data collection approaches include questionnaires, tests, peer reviews, self-reports, and audio-visual capture. These approaches have been foundational in understanding students' participation, interactions, behaviors, and learning progress [34]. However, these methods are susceptible to bias, which challenges accurate analysis and interpretation of results [34]. In recent years, learning analytics have shifted toward automated data collection, leveraging advancements in machine learning and deep learning algorithms [35]. This approach aims to mitigate bias, offering more objective insights into students' behavioral, cognitive, and learning patterns. In learning spaces, audio and video data are captured via strategically placed devices [36] or participants' devices [29], ensuring that high-quality data acquisition is essential for accurate processing and reliable results. Notably, these student data's capture, storage, sharing, and reuse have become increasingly prevalent with the surge of interest in educational technology [37]. Although automated multimodal data collection promises greater analytic precision, it simultaneously amplifies the scope, persistence, and granularity of surveillance within educational contexts. This shift from episodic assessment to continuous data capture

introduces a fundamental tension between analytic insight and informational self-determination.

While these advancements in learning analytics have undoubtedly enhanced learning experiences and outcomes, a pertinent concern arises regarding the potential invasiveness of these tools [38]. Additionally, several works have underscored the privacy and data protection issues prevalent in learning analytics [23,24]. Notably, the pervasive practice of capturing video and audio data in learning spaces has emerged as a transformative yet contentious innovation [39]. While providing valuable insights into student engagement and performance, it raises substantial concerns regarding privacy and data protection [26]. The privacy issues surrounding video recording in educational settings raised by Frank et al. [26], include loss of anonymity, risk of reputational damage to educators perceived as embodying suboptimal teaching practices, and breaches of promises regarding subject confidentiality. Across these studies, privacy concerns extend beyond the act of recording itself to the data lifecycle. That is, how it is stored, accessed, repurposed, and potentially linked beyond its original pedagogical context. This distinction highlights that perceived invasiveness arises not only from visibility during recording but also from uncertainty surrounding future data trajectories.

Research has shown significant privacy concerns regarding the sharing and reuse of video data generated from student class activities [40,26]. Frank et al. [26] reported that data producers and reusers were concerned with the risks associated with qualitative data, especially videos depicting classroom discussions and group work. They highlighted concerns regarding the privacy of the teachers and students featured in the videos. Therefore, while effective data gathering, sharing, and reuse are fundamental to multimodal learning analytic tools, a critical concern emerges regarding how students' data are handled, from storage to transmission, without compromising their privacy [41]. This concern intensifies when the participants in the data are easily identifiable, as is the case with video data [26], especially during sensitive topic discussions.

Beyond individual-level privacy concerns, video-based learning analytics systems operate within broader institutional governance and regulatory frameworks. In the United States, the collection and storage of student video data intersect with federal regulations such as the Family Educational Rights and Privacy Act (FERPA) and Institutional Review Board (IRB) oversight governing the research use of educational records. These frameworks shape institutional responsibilities regarding data access, consent, secondary use, and retention policies [42,43]. Particularly, emerging best practices in educational data governance emphasize principles such as data minimization, purpose limitation, role-based access control, and transparency in long-term data storage and reuse [44]. Prior work has argued that student trust in learning analytics systems depends not only on technical safeguards but also on clearly articulated governance structures and institutional accountability mechanisms [45]. However, even robust governance mechanisms do not automatically resolve hesitancy about participation in video-mediated discussions, particularly when students perceive ongoing identity exposure during interaction. Thus, privacy must be understood as both a regulatory condition and a situated experiential phenomenon. While the present study focuses on students' perceived comfort and willingness to allow video storage for analytics purposes, these perceptions are situated within this broader governance landscape, where institutional safeguards and policy structures influence trust in video-mediated learning environments.

2.3. Video-based MMLA tools: opportunities and challenges

MMLA tools embedded with video devices have become increasingly prevalent in capturing student interactions during class discussions [46]. These tools offer unprecedented opportunities for understanding student engagement, behavior, and learning processes. For example, Soloviev [47] proposed a system for continuous analysis of video data

collected from classroom cameras. It uses machine learning models to identify students' faces, recognize their emotions, and evaluate their engagement levels. Similarly, Woolf et al. [48] explored AI-assisted individualized learning, using computer vision to inform intelligent tutors and teachers about student affect, motivation, and performance. They argued that facial expression recognition is useful in identifying subtle differences when students ask for hints or fail to solve problems. Across these systems, a common assumption is that richer visual access to learners, particularly facial cues and affective signals, enables more precise modeling of engagement and adaptive instructional support. However, the analytic benefits of detailed visual capture are structurally intertwined with increased visibility of student identity, raising questions about how much observational granularity is pedagogically necessary versus experientially intrusive.

While video-based activities in classrooms have shown potential benefits, they also raise significant privacy concerns [49]. Studies on students' perceptions of video-based activities [50,51] have generally revealed positive attitudes toward these technologies, acknowledging their potential to foster engagement and collaboration. However, these studies also highlighted the need to address privacy concerns and establish clear guidelines for the appropriate use of video content. Régó et al. [52] explored ethical considerations surrounding care and privacy in online classroom environments. Their findings reveal that teachers and students are vulnerable to various ethical risks, including unwarranted exposure, insufficient privacy safeguards, and the overarching surveillance context inherent to online educational platforms. These concerns become particularly problematic when students are involved in video-recorded sensitive-topic discussions.

Importantly, much of this prior work evaluates student attitudes toward video use at a general level (e.g., acceptance, satisfaction, perceived usefulness), rather than examining how privacy perceptions influence expressive participation during high-stakes or sensitive discussions. As a result, a distinction emerges between positive endorsement of technology in principle and behavioral modulation under conditions of perceived identity exposure. Taken together, the literature on video-based MMLA tools reveals a central tension: systems designed to capture nuanced affective and interactional signals rely on sustained visual access to learners, yet this same visibility can recalibrate participation dynamics by heightening self-presentational awareness. While prior research has demonstrated the analytic power of video-based systems and separately documented privacy concerns, fewer studies have empirically examined how modifying visual representation itself may alter learners' expressive behavior within recorded interaction spaces. This unresolved tension between analytic richness and experiential privacy provides the conceptual bridge for the present study, which investigates whether a privacy-aware visual transformation can recalibrate participation conditions without eliminating the expressive cues upon which multimodal analytics depend.

2.4. Impact of visual privacy on student engagement

Various visual privacy protection techniques have been applied across diverse domains, ranging from learning space, ambient assisted living, and privacy-preserving smart rooms to video action recognition, safeguarding against malicious facial recognition models, and in-bed human pose estimation [53–57]. Researchers have proposed various methods for safeguarding visual privacy. These methods range from mere obfuscation, such as blurring, pixelation, and mosaicking of faces [27], to more sophisticated techniques, such as complete facial de-identification, which still preserves details that might be important for rich analytics [58,59,27]. Across these approaches, visual privacy techniques differ not only in the degree of identity alteration but also in their underlying assumptions about what constitutes “sufficient” privacy. Obfuscation-based methods prioritize concealment but often sacrifice expressive detail, whereas generative or stylized transformations seek to balance identity alteration with retention of facial geometry

relevant for downstream analysis. This reflects a broader methodological tension between maximizing anonymity and preserving analytics. However, visual de-identification techniques should not be equated with comprehensive privacy guarantees, as privacy encompasses contextual, relational, and perceptual dimensions beyond the removal of identifiable facial detail.

Padilla-López et al. [28] proposed diverse image privacy protection methods, including image redaction techniques, video and image encryption, face de-identification, object or person removal, and visual abstraction or object replacement. Maity et al. [60] explored the intersection between video analytics and privacy preservation, focusing on face deidentification and background blurring. Importantly, certain computer vision operations, such as person and action detection, do not necessarily require detailed facial features in a video. Nonetheless, the indiscriminate obfuscation of facial images can lead to significant image alteration and degradation [59]. Therefore, retaining facial features is valuable for subsequent visual analytics applications such as eye tracking, facial emotion detection, and joint attention. This enhances the analytical capabilities of visual data while supporting participants' perceived privacy in routine educational contexts.

Although considerable attention has been given to visual privacy and data protection regarding sensitive information, there has been limited substantive research on visual privacy protection and its impact on participants in a learning space. Few studies have explored the implications of visual privacy protection for students in a learning environment. For example, Li et al. [61] proposed a computer vision for position estimation (CVPE) approach, which focuses on collecting socio-spatial behavior data to unravel teamwork dynamics. The study incorporated a built-in privacy protection module that employs face masking with black boxes. However, the limitation of this approach lies in the difficulty of leveraging facial features for other multimodal analyses. Similarly, Geddani [62] explored promoting active learning by protecting student privacy via two face deidentification models (attGAN and Fawkes) to transform students' faces. Although the study successfully predicted students' engagement from detected emotions in the de-identified images, challenges arose because no significant difference was observed between the original and de-identified images.

Thus, these studies illustrate divergent strategies: some approaches prioritize stronger visual masking at the expense of expressive richness, while others retain structural facial information but prioritize analytic feasibility over learner perception. What remains comparatively underexplored is how privacy-preserving transformations (cartoonification) recalibrate socio-emotional participation in sensitive-topic dialogue, where identity exposure and expressive risk are heightened. Importantly, visual transformations such as cartoonification are more appropriately conceptualized as privacy-enhancing visualization strategies that reduce perceived identity exposure within routine educational contexts rather than as guarantees of adversarial anonymity. This distinction is critical in learning analytics settings, where preserving expressive facial dynamics for analytic interpretation must be balanced against students' perceptions of visibility and control over their digital representation.

2.5. The need for visual privacy in fostering students' participation and engagement in sensitive topic discussions

Lusk and Weinberg [19] reported that introducing controversial subjects often leads to uncomfortable silences or superficial responses from students, primarily due to their fear of negative repercussions. This reluctance also stems from privacy concerns about linking their opinions to their face and voice [10] when engaging openly with sensitive topics. This creates a significant barrier to fully realizing the benefits of classroom discussions and limits the quality of data collected in this context.

Researchers have proposed various teaching methods to address the challenges of discussing sensitive subjects. Macmanus et al. [63] advocated creating a welcoming and inclusive environment that enables

students to explore their ideas on controversial questions. Kramer [64] suggested discussion strategies such as the circle of understanding, student-led discussions, burning questions, and role-play activities to ensure meaningful learning. Steffensen et al. [65] explored the use of inquiry-based dialogue for investigating controversial topics with teenage students. While these pedagogical strategies aim to cultivate psychological safety and inclusive dialogue, they primarily address interpersonal and classroom climate factors rather than the technological conditions under which discussions are recorded and stored. In video-mediated contexts, concerns about identity traceability and future data reuse may persist even in well-facilitated environments. Thus, a distinction emerges between relational safety (fostered through instructional design) and representational safety (shaped by how learners' visual identities are captured and stored).

Furthermore, prior work on visual de-identification has largely evaluated technical feasibility, such as preserving facial landmarks or resisting recognition, without examining how transformed visual representations influence expressive participation in high-stakes dialogic settings. This leaves an underexplored intersection between visual privacy design and socio-emotional participation during sensitive discussions. This study aims to address this gap by investigating the impact of cartoonifying participants' video data (visual privacy) on their expressive participation in discussions of sensitive topics.

The unique aspect of this study lies in its exploration of whether the visual transformation of students' faces, while essential facial features are retained, reduces perceived identity exposure sufficiently to enable uninhibited expression of emotions and opinions in learning environments. To achieve this goal, we evaluated various machine-learning models to identify the most effective for facial privacy preservation in our study. Our primary criterion was selecting a model that could significantly alter facial appearance, creating a notable distinction from the original face while preserving essential facial features. This preservation of key features was crucial, as it can enable subsequent visual analytics applications such as eye tracking, facial emotion detection, and joint attention analysis. By maintaining this delicate balance between visual privacy and feature retention, we aim to preserve expressive facial structure while reducing identifiable detail, supporting potential analytic interpretability. In this sense, the intervention is positioned not as absolute anonymization but as a participation-sensitive design strategy that seeks to recalibrate perceived visibility while retaining expressive utility. The emphasis shifts from maximizing concealment to optimizing the privacy–utility balance in ways that meaningfully influence learner experience.

We consider a model by Maximov et al. [59], which leverages the generative power of conditional generative adversarial networks (CGANs) [58] to develop a CIA-GAN model. This model generates realistic anonymized images and videos while retaining necessary detection, recognition, or tracking features. However, a drawback of this model is its limited range of image styles and lack of significant differences between the original and anonymized faces. Another model considered was the Vtoonify framework [27], which adopted a novel approach to synthesizing high-resolution images or videos from low-resolution inputs. This model has a distinct advantage in offering multiple styles, including caricatures, cartoons, and animations. These diverse styles enable varying degrees of privacy, catering to different preferences and requirements. Notably, this model stands out for its ability to provide a more pronounced differentiation between the original and anonymized faces. We also used Py-Feat [66] to detect facial emotions in both the original images and their transformed Vtoonify versions, finding no remarkable differences in emotion between the two. Given the unique strengths and considerations, we adopted the Vtoonify framework.

In conclusion, this study aims to bridge the gap among visual privacy, video data collection, and student engagement by investigating the impact of face privacy through cartoonification on students' expressive participation during sensitive-topic discussions, using the

frequency of emotional tones. Additionally, it examines the impact of privacy-enhancing techniques on students' willingness to be recorded *during* challenging classroom discussions. This research contributes to the ongoing discourse on privacy, technology, and the socioemotional dynamics of learning environments. The findings from this study have the potential to inform the design of future educational technologies and pedagogical approaches, fostering more inclusive and productive learning environments that balance the benefits of data-driven insights with strengthened visual privacy safeguards.

3. Method

This exploratory study investigates how implementing video privacy on the BLINC platform increases students' willingness to participate in video-recorded activities, enhances their socio-emotional expression of candid opinions during challenging classroom dialogue, and preserves the benefits of video-based analytics. A multi-method approach was employed to address the research question, combining quantitative methods. The study involved students engaging in discussions about police brutality and racism, AI and racial bias, and the tensions between science and religion. These topics were selected for their sensitivity and controversy, providing an ideal context for examining the impact of privacy-enhancing features on student engagement. Tailored questions were designed to measure participants' engagement and emotional tone during discussions of these sensitive topics, with the cartoonification feature toggled on and off to enable comparative analysis. Details of participant pairing, topic assignment, and condition randomization are described in [Section 3.2](#).

The goal is to determine whether the cartoonification feature meaningfully reduces perceived identity exposure, supports participants' privacy, and fosters confidence in openly expressing their opinions. The frequency of emotional tones (sadness, joy, love, anger, fear, surprise, and concern) detected in transcripts was analyzed as an observable indicator of affective expressiveness during dialogue, interpreted alongside self-reported measures of confidence and willingness. Following the discussions, students completed a post-session survey that documented their overall experience and their willingness to have video data collected for analytics and research purposes. These operationalized constructs informed the analytic strategy and guided the examination of expressive participation and perceived privacy control in the study design. This methodological approach was selected for its capacity to offer a comprehensive view of the phenomenon under investigation. The combination of self-reported survey data and objective analysis of emotional tone in discussion transcripts enables triangulation of findings, thereby enhancing the reliability and validity of the results.

3.1. Cartoonification feature implementation

The facial cartoonification feature added to an existing learning analytic platform, BLINC, visually transformed participants' faces, rendering them distinctly different from their original appearance while retaining essential facial features that remain compatible with learning analytics interpretation, as shown in [\(Fig. 1\)](#). The cartoonification transformation preserves coarse facial structure, mouth and eye region movement, and temporal expressive dynamics commonly leveraged in affect-related analysis of learning interaction [\[27\]](#). Accordingly, we

interpret the transformed representations as structurally compatible with learning analytics methods that rely on expressive facial geometry. In this study, cartoonification is designed to reduce recognizable facial detail in routine educational contexts to support participants' perceptions of privacy and to examine its influence on expressive engagement and willingness to participate in future data use in learning analytics settings. The transformation targets reducing perceived identity exposure during recorded interactions rather than defending against sophisticated human or automated re-identification attacks. The primary privacy concern addressed is students' perceived visibility and identifiability by instructors, peers, and researchers during or after recorded discussions, particularly in sensitive-topic contexts. Accordingly, it should be interpreted as a privacy-enhancing visualization layer rather than as a formally validated guarantee of resistance to adversarial re-identification. In addition to visual privacy, the platform preserved participants' verbal interactions and allowed toggling of the cartoonification feature at the session level. The system did not modify audio signals or alter linguistic content, ensuring that transcript-based emotional analysis reflected authentic participant dialogue. Students participated in a discussion on a sensitive topic with the cartoonification feature activated or deactivated. The real-time capture of participants' video data occurs at 10-second intervals and is then converted into frames. Within each frame, the facial detection component was implemented using a pre-trained CNN face detector, Dlib (CUDA-enabled), to improve detection reliability compared with a CPU-only configuration [\[16\]](#).

In rare cases where facial landmarks could not be detected due to limited facial visibility (e.g., extreme head pose or partial occlusion), the affected frame was skipped rather than transformed. Because these events were infrequent, skipping frames did not materially affect real-time operation or the consistency of the cartoonification experience across sessions. To achieve the desired cartoonification effect, image-based style transfer is subsequently applied to the predicted facial landmarks. This transformation is facilitated by leveraging the VToonify [\[27\]](#) pre-trained machine learning model. A single style transformation was applied to visual rendering across experimental conditions, ensuring that any observed differences were attributable to the presence or absence of the cartoonification feature rather than to variations in model configuration. The real-time streaming of these transformed frames to the dashboard completes the process, enabling continuous monitoring of the transformed facial representations during discussions on the BLINC platform. This meticulous methodology ensures participants' privacy while maintaining the feasibility of real-time interaction and engagement within the learning environment.

3.2. Experimental design

The experimental design used the BLINC platform, which included an optional cartoonification feature for video-based discussions. A total of 36 participants were recruited and randomly paired to form 18 pairs, ensuring a balanced and varied dyad composition. This random pairing approach minimized selection bias and eliminated potential opinion bias while promoting openness among peers. Sensitive discussion topics were randomly assigned to groups to prevent systematic bias that could arise if certain groups consistently discussed a particular topic. To enable a direct comparison between conditions, the 18 groups were



Fig. 1. Samples of original images and the cartoonized versions.

randomly assigned to participate with the cartoonification feature either activated or deactivated, yielding 9 groups per condition. Participants in the cartoon-on condition were aware of the feature during the discussion, as it was visibly active on their video feed; participants in the cartoon-off condition were not informed about the feature until the end of the session, when they watched a brief demo. This randomized assignment strategy was deliberately chosen to mitigate potential emotional biases that might arise if participants were (aware) exposed to both conditions. To ensure that each group experienced only one condition, we sought to elicit authentic emotional responses to the discussion topics without prior exposure to the alternative condition. However, at the end of the session, participants in groups where the cartoonification feature was deactivated were shown a demonstration of how the feature works to assist them in accurately completing the survey.

The participants were informed that discussions would be recorded solely for analysis in the context of this study, ensuring ethical compliance and allowing for informed consent. While participants consented to have their video recorded and stored for further analysis in this study, the survey further explored whether the cartoonification feature strengthened their willingness to allow their video data to be collected for future educational research and learning analytics. All discussions were audio- and video-recorded and transcribed, providing rich data for detailed analysis of emotional tone. At the end of the discussion session, all participants completed the first, second, and fifth survey questions, and then responded to either the third or fourth question, depending on whether cartoonification was activated or deactivated during the discussion. Moreover, all groups interacted within the same platform interface and discussion structure to ensure consistency across conditions.

Therefore, the cartoonification serves as a privacy-aware participation layer, reducing identifiable visual details while retaining expressive facial dynamics compatible with learning analytics interpretation. The experimental protocol evaluates the human-centered implications of this expression-preserving privacy design rather than benchmarking downstream visual analytic accuracy. This carefully crafted design allowed us to systematically examine the impact of the cartoonification feature on participants' comfort and confidence level in expressing their opinions during sensitive topic discussions.

3.3. Participant selection

The composition of our participant pool was carefully considered to enhance the validity and relevance of our study. We selected 36 students from a private university in the United States to provide a diverse yet manageable sample size. The inclusion of 28 undergraduates, 2 masters, and 6 PhD. students allowed us to capture perspectives across different levels of academic experience and maturity, which is particularly valuable when discussing sensitive topics that may be perceived differently based on educational background. The gender distribution consisted of 22 males and 14 females, although not perfectly balanced, which offered insights from both gender perspectives. The racial backgrounds of the participants were 13 Black or African American participants, 14 Asians, 8 whites, and 1 Hispanic or Latino individual.

This racial diversity provided a key strength to our sample, as the diversity of opinions and frequency of emotional tones expressed provided an open perspective on discussions involving sensitive topics such as police brutality and racism, AI and racial bias, and the tensions between science and religion. While the sample size for some groups (e.g., Hispanic or Latino) was limited, the overall diversity of our participant pool strengthens the ecological validity of our study, making our findings more representative and applicable to the diverse student populations found in many higher education settings in the United States. Following the discussion activity, all participants completed a concluding survey and received compensation of a \$20 gift card.

This study received approval from the Institutional Review Board.

Prior to participation, students provided informed consent and were informed that video recordings would be used for research analysis within the scope of this study. Because discussions involved sensitive topics, participants were reminded that they could skip questions, pause participation, or withdraw at any time without penalty. Video and transcript data were stored on secure, access-controlled institutional servers and were accessible only to authorized members of the research team for analysis purposes.

3.4. Data collection

In this study, each group of participants engaged in a 20-minute discussion on a randomly assigned sensitive topic, using the BLINC platform, with a microphone and video enabled. Cartoonification was activated in 9 of the 18 groups. Dyads were randomly paired to minimize familiarity-based bias that could influence participants' willingness to express their opinions during discussion. All discussion sessions followed a standardized procedure. Each dyad was assigned one discussion topic selected at random from three sensitive themes. A standardized set of guiding questions associated with that topic was provided to initiate dialogue while allowing natural conversational flow. No active facilitation or instructor intervention occurred during the discussions to minimize external influence on participants' emotional expression. Participants joined remotely using their personal devices in a synchronous online setting [67].

The video camera captured participants' portrait postures to obtain high-quality facial data. Moreover, Participants were briefly informed that maintaining visible facial orientation toward the camera would help ensure consistent application of the cartoonification effect. This guidance was provided solely to support the technical reliability of facial detection and did not impose constraints on conversational content or interactional dynamics. Standardized prompts, identical discussion duration, and consistent platform configuration were implemented to support comparable measurement conditions across experimental groups. During the discussion, the BLINC platform captured participants' spoken contributions as audio data and generated discussion transcripts in real time. These transcripts were subsequently exported for analysis of emotional tone as an observable indicator of affective expressiveness during dialogue.

Following the discussion activity, all 36 participants completed a post-session survey that served as a structured measurement instrument to capture perceived comfort with video recording, confidence in expressing opinions during sensitive discussions, and willingness to use data in the future. Participants who engaged with cartoonification responded to Questions 1, 2, 3, and 5, whereas those in the non-cartoonification condition responded to Questions 1, 2, 4, and 5. Although wording differed to reflect participants' assigned condition, Questions 3 and 4 were parallel items assessing the same underlying construct, confidence in expressing opinions during sensitive dialogue.

Q1 How comfortable are you having your video recorded during class discussion activity?

Q2 How comfortable are you having your video recorded during out-of-class discussion activity?

Q3 How confident were you in expressing your opinion and ideas during the sensitive topic with the cartoon feature turned on?

Q4 How confident were you in expressing your opinion and ideas on a sensitive topic without the cartoon feature?

Q5 How willing are you to allow your video to be stored for analytics in future video-based class activities?

3.5. Model development

We developed a robust emotion classification model by combining multiple emotion datasets [68–70] and carefully aligning the emotional labels with the tone of expression in our transcripts. From the dataset

[69], we selected text emotions with labels (sadness, joy, love, anger, fear, surprise) following the Ekman taxonomy [71]. However, recognizing that sensitive discussions often elicit complex psychological states, we expanded our taxonomy to include concern, a category encompassing worry and anxiety, as defined by Saravia et al. [70] and Kleinberg et al. [68]. We ensured that the addition of this category was theoretically grounded and relevant to the context of socio-emotional expressiveness during discussions of controversial topics. Table 1.

The 7 emotion labels were encoded as integers from 0 to 6 using a scikit-learn LabelEncoder. The TensorFlow Keras one-hot function was used to encode the sentence. Our final dataset comprises 428,404 instances, which are split into training (80%) and validation (20%) sets. We implemented and trained a sequential multilayer convolutional neural network (CNN) classifier using TensorFlow Keras, achieving a training accuracy of 89% and a validation accuracy of 88%. The emotion classification model was applied consistently to all transcripts without condition-specific adjustments. The model's performance matrix was computed as shown in Table 2. Overall, the model demonstrated strong performance across all emotion classifications.

3.6. Data analysis

Our data analysis approach aimed to comprehensively evaluate the influence of video cartoonification on students' socio-emotional participation during discussions of sensitive topics, using frequency counts of emotional tones. The influence on students' willingness to be recorded in future challenging classroom discourse should be examined for analytics and research purposes. We employed a multi-method quantitative analysis strategy, combining emotion-coded transcripts with survey responses. This approach allowed us to triangulate our findings and provide a nuanced understanding of the effectiveness of cartoon features.

3.6.1. Transcript analysis

We analyzed participant transcripts using the machine-learning text-emotion classifier to quantify the emotional tone of the discussion transcripts. This method supports our objective of assessing the impact of cartoonification on students' emotional expressive engagement by analyzing the frequency with which emotions are expressed during discussions of sensitive topics. All recorded discussion sessions were transcribed using an automated speech-to-text system integrated within the BLINC platform. Subsequently, the transcripts were segmented at the utterance level by speaker turn. Furthermore, Sentence-level segmentation was performed using the NLTK library. Prior to analysis, text preprocessing included lowercasing, removal of non-linguistic artifacts, and normalization of punctuation. We then prepared an Excel spreadsheet with one column for each group, each row containing a different sentence transcript. The trained model was then used to predict the emotional tone of the discussion transcripts. The predictions for each sentence were aggregated into counts, providing insights into the frequency distribution of emotions across the various groups. Model performance metrics (validation accuracy and class-level precision, recall, and F1 scores) ensured analytical reliability, and aggregating emotion frequencies at the group level reduced the influence of isolated classification errors.

Table 1

Training and validation dataset distribution.

Emotions	Samples	Train (80%)	Validation (20%)
sadness	121,187	96,950	24,237
joy	141,067	112,854	28,213
love	34,554	27,643	6911
anger	57,317	45,854	11,463
fear	47,712	38,170	9542
surprise	19,562	15,650	3912
concern	28,166	22,533	5633
Total	449,565	359,654	89,911

Table 2

Classification metrics.

Emotions	Precision	Recall	F1-Score
sadness	0.95	0.95	0.95
joy	0.92	0.93	0.92
love	0.78	0.84	0.81
anger	0.96	0.87	0.91
fear	0.83	0.93	0.88
surprise	0.87	0.51	0.64
concern	0.80	0.91	0.85

Given the non-normal distribution of our data (confirmed by the Shapiro-Wilk test), we employed the Mann-Whitney U test to assess statistical differences and effect sizes in emotional tone expression between groups with and without cartoonification. The analytic pipeline focused on socio-emotional indicators derived from transcript-based frequencies of emotional tone, rather than behavioral interaction metrics. Thus, this analysis directly addressed our research question by quantifying the impact of cartoonification on students' expressive participation during discussions of sensitive topics using the frequency of detected emotional tones.

3.6.2. Survey analysis

We conducted a post-discussion survey to complement transcript-based emotional tone analysis and capture participants' subjective perceptions of comfort, confidence, and willingness toward future data use. The survey consisted of targeted single-item measures aligned with the study's research questions and grounded in Privacy Regulation Theory [31,32], which conceptualizes privacy as a dynamic balance between disclosure and perceived control over personal information. Perceived comfort with video recording (Q1, Q2; extremely uncomfortable to extremely comfortable), and confidence in expressing opinions during sensitive discussions (Q3, Q4; very low confidence to very high confidence) were assessed using a 5-point Likert-type scale, allowing participants to indicate gradations in their attitudes. Also, Likert-scale responses were treated as ordinal indicators; means and standard deviations are reported for descriptive comparison, while interpretations remain cautious given the limited scale range [72,73]. In contrast, willingness to allow video data storage for future analytics (Q5) was framed as a decision-oriented judgment and measured using a reduced categorical response format. Survey design literature suggests that fewer response options may be appropriate when eliciting consent-style or forced-choice decisions, as simplified categories help reduce ambiguity and support clearer interpretation of participants' preferences [74].

Condition-specific wording was used for Q3 and Q4 to reflect whether participants experienced cartoonified or non-cartoonified video, while the underlying construct, confidence in expression during sensitive dialogue, remained consistent across conditions. Because the survey items functioned as targeted indicators rather than components of a multi-item psychometric scale, internal consistency reliability analyses were not performed. Instead, the items were analyzed descriptively to capture participants' perceived experiences and decision-making tendencies within the experimental context. For each question, we computed descriptive statistics including mean, standard deviation, and variance to examine patterns across conditions. The survey findings were interpreted alongside transcript-derived emotional tone analysis to provide a complementary perspective on expressive engagement. This integrated multi-source quantitative approach enabled comparison between observable interactional signals and participants' self-reported perceptions, supporting a richer understanding of how cartoonification influences participation in sensitive discussions.

4. Results

This section presents the findings from our experimental

investigation into how AI-based visual privacy, implemented through real-time video cartoonification, influences students' dispositions toward expressing their opinions during sensitive-topic discussions and their willingness to have video data recorded and stored for learning analytics purposes. The results are organized in alignment with the research questions: first, we examine the emotional tone analysis to assess how cartoonification affects the openness and emotional expressiveness of student contributions; second, we analyze survey responses to evaluate their effect on students' comfort with being recorded and their willingness to allow the preservation of learning analytics data. Together, these findings provide insight into both the participation-related and data-quality-related implications of introducing visual privacy measures in video-based multimodal learning analytics.

Our study revealed that the frequency of emotional tone (Section 4.1) reflects the significant impact of the cartoonification feature on participants' expressive engagement during discussions of sensitive topics. Emotional tone analysis showed consistently higher levels of emotional expression when cartoonification was activated across all discussion topics, with concern as the most prevalent emotion. Notably, across the three topics, this effect was statistically significant and demonstrated a large effect size ($p = 0.001$, $U = 7$, $r = 0.7$; $N_{\text{on}} = 9$ dyads, $N_{\text{off}} = 9$ dyads). These findings were further supported by survey responses (Section 4.2), which indicated that students' self-reported comfort and willingness to have their data recorded and stored aligned with the observed changes in emotional expressiveness when the cartoonification feature was active or inactive.

Taken together, the results suggest that privacy-enhancing features such as cartoonification foster more open, emotionally expressive, and confident participation in educational discussions on sensitive topics, potentially creating more engaging and productive learning environments. In the following sections, we present detailed analyses of these findings, beginning with an analysis of emotional tone, followed by

survey-based responses.

4.1. Emotion analysis results

To address the first research question, how cartoonification affects the openness and emotional expressiveness of student contributions, we examined the distribution and frequency of emotional tone expressions in the discussion transcripts. As described in Section 3.2, participants in the cartoon-on condition could see their own video feed with the cartoonification effect applied in real time, whereas those in the cartoon-off condition did not see the feature until after the discussion concluded. Our objective was to determine whether the cartoonification feature increased participants' emotions expressive engagement during discussions of sensitive topics. We began by assessing the distribution of emotional tone scores using the Shapiro–Wilk test for normality. The test indicated that the data were not normally distributed, violating the assumptions for parametric testing. Consequently, we used the Mann–Whitney U test, which does not assume normality, to compare emotional tone levels between the cartoon-on and cartoon-off conditions. All statistical analyses were conducted at the dyad level, where emotion predictions were aggregated within each discussion pair to produce a single observation per dyad ($N = 18$ dyads; 9 cartoon-on, 9 cartoon-off), thereby preserving independence despite the conversational structure. While statistical comparisons were performed at the dyad level, interpretations focus on participants' expressive behaviors emerging through collaborative interaction.

Across all topics combined (Fig. 2d), dyads in the cartoon-on condition exhibited higher aggregated concern expression than dyads in the cartoon-off condition ($U = 7$, $p = 0.001$, $r = 0.7$; $N_{\text{on}} = 9$, $N_{\text{off}} = 9$). Median concern counts indicated a clear distributional shift, with cartoon-on dyads exhibiting higher values (Median = 30, IQR = 23–34) compared to cartoon-off dyads (Median = 17, IQR = 17–19). The

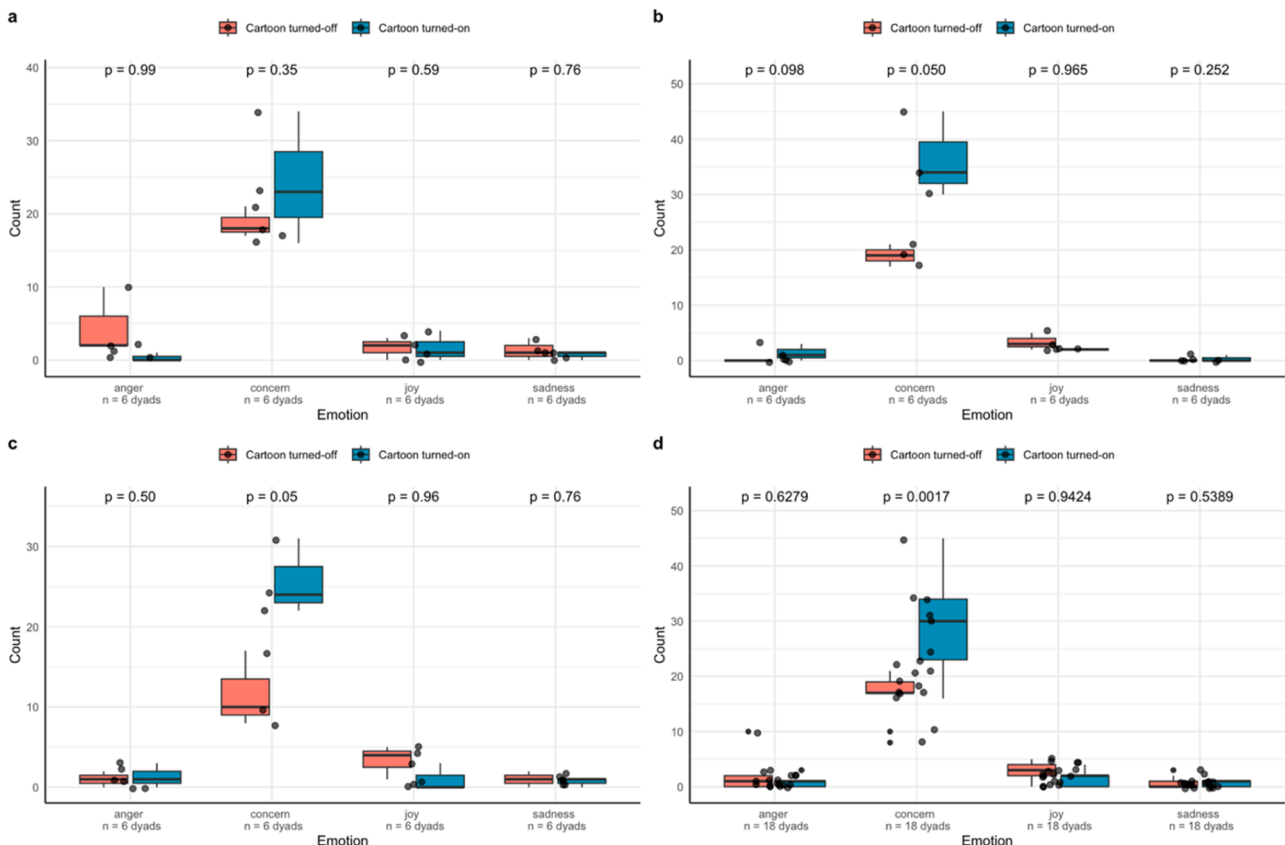


Fig. 2. Emotion frequency distribution across discussion topics with and without cartoonification.

median difference was 13 (95% bootstrap confidence interval [4, 17]), indicating that the observed effect remained positive across resampled dyad distributions. Jittered dyad-level data points (Fig. 2d) further demonstrate that the effect was not driven by a single extreme observation, and a sensitivity analysis excluding the highest-value dyad produced a similar pattern of results ($U = 6$, $p = 0.0036$, $r = 0.7$; $N_{on} = 8$, $N_{off} = 8$), supporting the robustness of the finding (see Appendix A for sensitivity analysis results). Taken together, these dyad-level results suggest that participants engaged in more open and emotionally expressive dialogue when visual privacy was enabled through cartoonification. The prominence of concern emotion across topics likely reflects the sensitive and controversial nature of the discussion prompts, within which the privacy-enhancing feature appears to have supported more confident emotional expression.

Topic-level analyses were conducted to provide contextual insight into how emotional expression varied across discussion contexts, given the limited number of dyads per topic ($n = 6$; 3 per condition). In discussions on police brutality and racism (Fig. 2a), no statistically significant difference was observed between conditions ($U = 3$, $p = 0.35$, $r = 0.3$; $N_{on} = 3$, $N_{off} = 3$). Concern expression showed greater dispersion in the cartoon-on condition (Median = 23, IQR = 19.5–28.5) compared to cartoon-off dyads (Median = 18, IQR = 17.5–19.5). The smaller median difference and larger variability suggest heterogeneous emotional engagement across dyads, which reduced statistical separation despite elevated concern expression. These patterns may reflect the emotionally charged nature of the topic, which can elicit strong responses regardless of visual privacy interventions [75–77].

Dyads discussing the conflict between science and religion (Fig. 2b) exhibited higher concern expression when cartoonification was activated ($U = 3$, $p \leq 0.05$, $r = 0.8$; $N_{on} = 3$, $N_{off} = 3$). Median concern values increased from 19 (IQR = 18–20) in the cartoon-off condition to 34 (IQR = 32–39.5) in the cartoon-on condition, indicating a strong distributional shift toward more expressive engagement. Similarly, discussions on AI and racial bias (Fig. 2c) showed a significant increase in concern expression with cartoonification enabled ($U = 0$, $p \leq 0.05$, $r = 0.8$; $N_{on} = 3$, $N_{off} = 3$). Cartoon-on dyads displayed a higher median concern count (Median = 24, IQR = 23–27.5) relative to cartoon-off dyads (Median = 10, IQR = 9–13.5), suggesting clearer separation between conditions compared to the police brutality topic.

Across topics, these patterns suggest that visual privacy may interact with contextual factors influencing emotional engagement, although the small number of dyads per topic precludes formal interaction modeling. Furthermore, no consistent differences in anger, joy, or sadness were observed across topics. Median distributions for these emotions largely overlapped across conditions, indicating that the most consistent effect of cartoonification was specific to concern-related emotional expression rather than a general increase in emotional output. Consequently, across analyses, dyads participating with cartoonification activated exhibited higher levels of concern expression, suggesting that participants may have felt more comfortable sharing emotionally nuanced perspectives when visual privacy was present. The consistency of this pattern across topics and in the combined analysis reinforces the conclusion that the cartoonification feature supported more expressive and confident participation during discussions of sensitive issues.

4.2. Survey results

The second research question focused on how cartoonification influences participants' comfort with video recording, their willingness to allow the preservation of learning analytics data, and their confidence in expressing opinions. To address this, we analyzed survey responses collected after the discussions. Survey responses were analyzed primarily through response distributions. Because the items were measured on a limited Likert-type scale, they were treated as ordinal indicators; means and standard deviations are reported for descriptive comparison only and are interpreted cautiously rather than as precise interval-level

estimates [72,73]. This result provided a complementary perspective to the emotional tone results (Section 4.1), allowing us to interpret whether participants' perceptions align with the observed changes in emotional expressiveness during discussions. By triangulating behavioral and attitudinal data, we provide a more comprehensive understanding of how visual privacy measures affect student participation and data retention preferences in sensitive learning contexts.

Fig. 3 presents an overview of the response distribution from the survey questions, revealing several noteworthy insights into the influence of cartoonification on participants' comfort level in being recorded, willingness to have recorded video data stored, and confidence level in expressing opinions during discussions. Regarding comfort level when the video was recorded (Q1 and Q2), we observed a slight preference for in-class settings over out-of-class activities. More participants (41.67%) expressed discomfort with video capture during out-of-class activities than during in-class activities (30.56%). Conversely, a greater percentage of participants reported extreme comfort with in-class video capture (16.67%) than with out-of-class capture (11.11%). Interestingly, an equal percentage of participants expressed neutrality about video capture in both settings. Overall, a majority of the participants (57.28%) indicated comfort with in-class video capture. However, the mean ratings, Table 3, for both in-class (Q1, mean=3.28, SD=1.22) and out-of-class (Q2, mean=3.11, SD=1.15) video capture fell within the *neutral* range but close to the *comfortable* range. Altogether, this suggests a fair level of comfort among participants regarding video capture in either setting.

The impact of the cartoon feature on participants' confidence in expressing their opinions was particularly pronounced (Q3 and Q4). With the cartoon feature activated, a substantial percentage of participants (50% and 11.11%) reported *High confidence* in expressing their opinions. In contrast, only a small proportion (11.11% and 5.56%) expressed *low confidence* under these conditions.

The opposite trend was observed with groups that participated when the cartoonification feature was deactivated. More participants (38.88% and 16.67%) reported *lower confidence* than *high confidence* (16.67% and 5.56%). Further supporting this trend, the mean rating for confidence with cartoonification activated (mean = 3.5, SD = 1.01) leaned toward *High Confidence*. In contrast, the mean rating without cartoonification (mean = 2.56, SD = 1.12) indicated only *low confidence*. This disparity suggests a positive association between the cartoonification feature and participants' confidence levels during discussions. Analyzing the responses to Q5 reveals compelling insights into participants' perspectives on the influence of the cartoonification feature on their willingness to have video data collected. A substantial proportion (41.67% and 44.44%) of participants are *very willing* and *moderately willing* to have their video data stored, while a very small percentage, 13.89%, are *not willing* to have their video data stored. These response distributions suggest that many participants perceived the cartoon feature as supporting willingness to have their data stored.

5. Discussion

This study sought to address two key challenges in video-based multimodal learning analytics (MMLA): students' willingness to be recorded for data collection and their engagement in sensitive topic discussions while maintaining the analytical value of video data. Guided by Privacy Regulation Theory [31,32] and Social Constructivism [33], our findings demonstrate that visual privacy through cartoonification plays a crucial role in reducing privacy concerns, fostering candid expressions, and increasing student participation in video-recorded discussions. The results both support and extend existing theory by showing that privacy-aware visual modifications can shape not only students' willingness to be observed but also the quality of dialogic engagement on sensitive issues. Furthermore, these results contribute to existing research on privacy in learning analytics [23,24] by providing empirical evidence that privacy-enhancing interventions can counteract

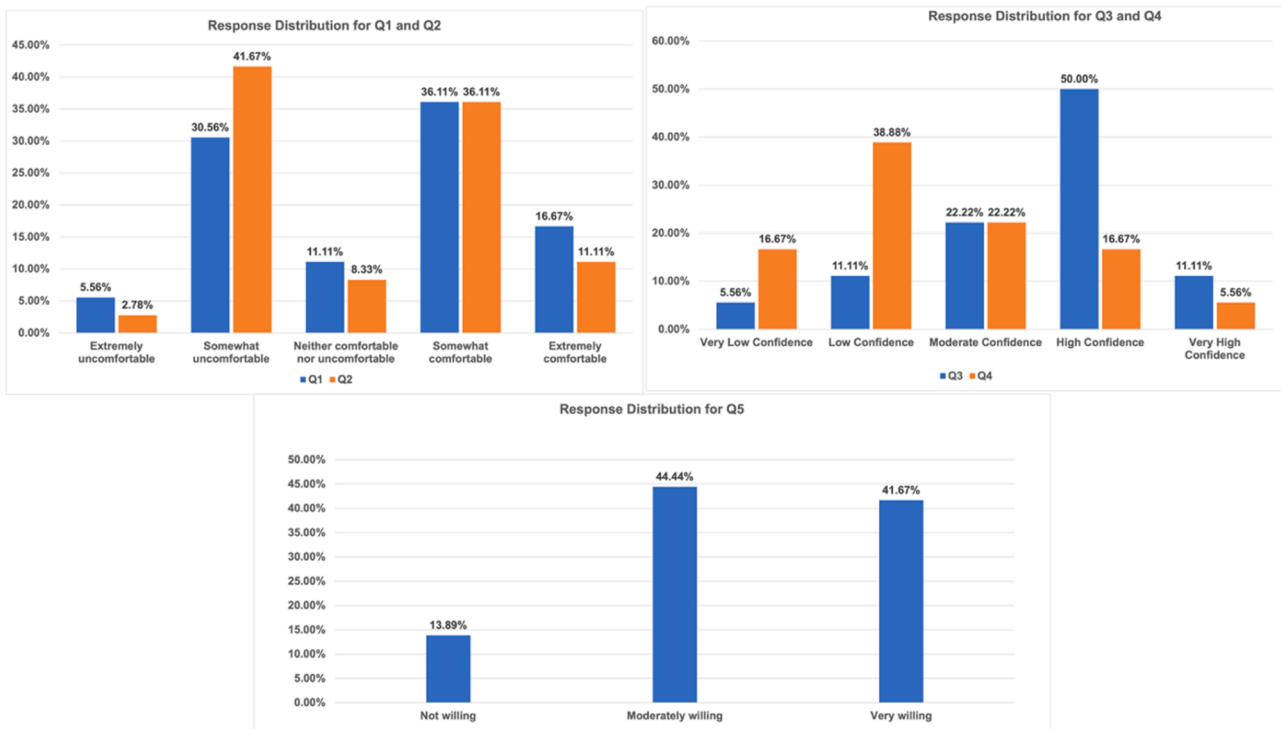


Fig. 3. Response distribution based on the survey questions.

Table 3
Statistical result for survey responses analysis.

Questions	Likert Scale	Mean	Std Deviation	Variance	Count
Q1*	5	3.28	1.22	1.48	36
Q2*	5	3.11	1.15	1.32	36
Q3 [#]	5	3.5	1.01	1.03	18
Q4 [#]	5	2.56	1.12	1.25	18
Q5 [~]	3	2.28	0.69	0.48	36

* 1.00–1.80(Extremely uncomfortable), 1.81–2.60(Somewhat uncomfortable), 2.61–3.40(Neutral), 3.41–4.20(Somewhat comfortable), 4.21–5.00 (Extremely comfortable).

[#] 1.00–1.80(Very Low Confidence), 1.81–2.60(Low Confidence), 2.61–3.40 (Moderate Confidence), 3.41–4.20(High Confidence) 4.21–5.00(Very High Confidence).

[~] 1.0–1.66 (Not Willing), 1.67–2.33 (Moderately Willing), 2.34–3.0 (Very Willing).

students' reluctance to participate in video-recorded learning activities [25,26]. This discussion synthesizes our findings to address the research questions and explore their broader implications for privacy, student engagement, and the future of video-based learning analytics. While Sections 5.1 and 5.2 correspond directly to the two research questions, we distinguish between (a) the pedagogical value of expressive participation in discussion and (b) the implications of recording for learning analytics, as these represent related but conceptually distinct outcomes of cartoonification.

5.1. Influence of cartoonification on willingness to record and store video data

A major concern surrounding video-based learning analytics is that students fear their recorded data may be stored, shared, or misused beyond the original classroom context [25,24]. Because participation in video-mediated discussion is closely tied to learners' perceptions of privacy and control, attitudes toward recording can shape students' willingness to engage in collaborative dialogue. Our study suggests that

these concerns persist, as reflected in students' generally neutral stance toward video capture. At the same time, the response distributions indicate greater reported willingness to have video data stored when cartoonification was enabled, suggesting that privacy-enhancing visual representations may help reduce participation barriers associated with identity exposure. Specifically, 41.67% of participants reported being very willing, and an additional 44.44% moderately willing to have their video data stored under cartoonification. These patterns align with research indicating that students are not necessarily opposed to video-based analytics itself but are often concerned about the visibility of personal identity in recorded interactions [22,23].

Additionally, descriptive summary statistics (mean = 2.28 on a 3-point Likert scale) were consistent with the observed response distribution, indicating a modest shift toward willingness to have video data stored. While the effect size remains moderate and should be interpreted cautiously, this pattern suggests that perceived visual privacy may support conditions under which students feel more comfortable participating in recorded learning environments [78]. Within a pedagogical context, increased willingness to be recorded may lower participation thresholds, potentially enabling more open dialogue during sensitive discussions.

Beyond pedagogical participation, these findings also highlight broader implications for the design and governance of video-based learning analytics systems. The results suggest that while privacy interventions such as cartoonification may not completely eliminate concerns about data storage, they can foster greater perceived control over one's digital representation. From a Privacy Regulation Theory perspective [31,32], the increased willingness to permit video storage when cartoonification was enabled suggests that participants adjusted disclosure boundaries in response to reduced perceived identity exposure, illustrating boundary-regulation processes rather than a fixed comfort threshold toward recording. Similarly, Zhang and Sundar [79] found that customizable privacy features can increase users' sense of agency and improve overall technology experiences. Although this research examined a single cartoonification style, the results suggest that offering multiple visual privacy transformations could further

support user agency in future MMLA systems.

Our findings also reinforce that students exhibit a cautious stance toward video recording, particularly in out-of-class settings. The survey results showed that a greater proportion of participants (41.67%) reported discomfort with video capture outside structured classroom settings compared to in-class activities (30.56%). Conversely, slightly more participants expressed extreme comfort with in-class recording (16.67%) than with out-of-class recording (11.11%). These patterns suggest that students may perceive greater privacy risks in unstructured environments, where expectations around data use and institutional oversight are less clear. These responses suggest that participants dynamically recalibrated disclosure boundaries across contexts, indicating that boundary management within Privacy Regulation Theory may operate not only at the individual level but also in relation to institutional cues and situational trust during recorded interaction. This interpretation aligns with prior studies showing heightened privacy concerns when video recording extends beyond clearly defined learning contexts [1,80].

In summary, while video-based analytics offer powerful tools for understanding student engagement and learning processes, their effectiveness is contingent on students' willingness to participate. This research suggests that privacy-preserving interventions such as cartoonification can encourage greater engagement in video-recorded activities by reducing concerns about long-term identity exposure. However, our findings also suggest the need for more comprehensive privacy policies that integrate multiple protection approaches to increase students' trust in video-based classroom activities.

5.2. Impact of cartoonification on student socio-emotional expressive participation

Our study provides key insights into the role of visual-privacy-enhancing features, specifically cartoonification, in fostering expressive participation during sensitive topic discussions in educational settings. The findings suggest that the visual privacy afforded by cartoonification supported more open emotional expression during collaborative dialogue. Participants who interacted through the cartoonified interface appeared more willing to articulate concern-related perspectives, consistent with theories of psychological safety in mediated collaboration. Because participants in the cartoon-on condition experienced the feature during discussion, whereas the cartoon-off group did not, initial novelty or expectation effects may have contributed to engagement patterns. However, the consistency of concern-related expression across topics suggests that perceived visual privacy likely played a substantive role in shaping expressive participation. These findings suggest that visual abstraction altered how participants regulated self-presentation during interaction, linking identity boundary management (Privacy Regulation Theory) with the discourse processes through which meaning is co-constructed in social constructivist learning environments.

Importantly, concern-related expression is not interpreted as a direct proxy for confidence or empowerment. Rather, concern may capture multiple forms of emotional engagement, including tension, ethical reflection, or discomfort, within discussions of sensitive topics. Accordingly, the findings are interpreted within a discourse-oriented framework emphasizing expressive participation rather than generalized positive affect. The selective increase in concern, rather than joy, sadness, or anger, suggests that cartoonification did not broadly amplify emotional intensity but may have supported reflective engagement with complex social issues. In dialogue involving sensitive topics, concern expressions often function as signals of ethical positioning or thoughtful participation, indicating that visual privacy may facilitate socially engaged expression rather than merely increasing affective output [81].

Building on this interpretive framing, the emotional tone analysis revealed a consistent shift in the distribution of concern-related expressions across discussion topics when cartoonification was active.

Rather than indicating a simple increase in emotional intensity, this pattern suggests that students may have felt more comfortable articulating nuanced perspectives when their visual identity was abstracted. Within a social constructivist framework [33], the shift toward concern-related expression may reflect increased externalization of ethical reasoning and perspective negotiation, suggesting that perceived visual safety enabled participants to engage more actively in the co-construction of meaning during sensitive dialogue [82,83].

Our findings align with and extend existing research on privacy, student engagement, and technology-assisted learning. Prior studies have emphasized that privacy awareness plays a critical role in shaping students' willingness to engage in discussions, particularly in recorded settings [83]. Our study extends this argument by providing empirical evidence that not only does privacy awareness impact engagement, but that actively integrating privacy-preserving features, such as cartoonification, significantly increases student participation and emotional expressiveness. This suggests that privacy-enhancing interventions are not merely perceived as beneficial but actively shape student behavior in learning environments. Furthermore, our findings extend social constructivist theory by operationalizing "safe space" not only as a socio-relational norm but also as a technological affordance through privacy-preserving visualization to facilitate the co-construction of knowledge during sensitive topic discussions.

Beyond pedagogical participation, these findings also raise important considerations for the design of recording-mediated learning analytics systems. Fang and Tse [84] demonstrated that addressing privacy concerns in AI-powered learning applications positively influenced class engagement and emotional expression, particularly when students felt their data was secure. Our study corroborates this perspective while making a novel contribution: unlike prior research focusing on privacy concerns in text-based or online learning environments, we examine video-based learning analytics, where concerns about identity exposure are heightened. Our results demonstrate that students' expressive participation alongside changes in perceived confidence ratings (mean shift from 2.56 to 3.5, $p = 0.001$), suggest that visual privacy representations may influence how learners navigate recorded interaction spaces.

Furthermore, the findings challenge overly restrictive de-identification methods that prioritize complete obliteration-anonymity at the expense of analytical value. Unlike Brkić et al. [85], whose facial transformation approach maximizes anonymity by removing nonverbal cues, our method retains key facial features commonly used in engagement-related analytical approaches. This balance between privacy and analytic utility highlights an ongoing design tension within video-based systems: safeguarding identity while maintaining interactional richness. These results highlight a sociotechnical tension that extends social constructivist perspectives by illustrating how technological affordances mediate the balance between identity protection and interactional visibility during collaborative meaning-making. While cartoonification may reduce identity exposure, it also reconfigures social signaling within collaborative environments, introducing potential trade-offs between perceived safety and accountability. This distinction is particularly important for learning analytics tools that rely on nonverbal cues, such as emotion recognition and engagement tracking [47,48].

5.3. Implications for privacy, learning analytics, and educational research

The implications of our findings are far-reaching and significant for the future of education, particularly in increasingly digital and diverse learning environments. Our results suggest that implementing privacy-enhancing features such as cartoonification can significantly improve student engagement in discussions on sensitive or controversial topics. This has important implications for fields such as social sciences, political science, and ethics education, where open dialogue on challenging subjects is crucial for learning. From an analytics perspective, by

demonstrating that visual privacy can be enhanced while retaining facial structures relevant to analytic interpretation, our study points to a future in which MMLA tools can balance student privacy concerns with the benefits of data-driven insights. However, for more robust privacy preservation, institutions must combine privacy-enhancing technologies, such as cartoonification, with clear policies on data handling, informed consent processes, student-centered customizable privacy settings, and transparency in data usage to ensure student trust in video-based learning analytics.

From a theoretical standpoint, our study shows that Privacy Regulation Theory (PRT) and Social Constructivism can be jointly leveraged to design MMLA systems that respect privacy while sustaining social learning. Cartoonification appears to operationalize boundary regulation by reducing perceived identity exposure (Privacy Regulation Theory), thereby lowering self-monitoring demands during recorded interaction and facilitating the collaborative externalization of ideas central to social constructivist learning processes. Practically, these findings suggest that privacy-enhancing features should be positioned not as add-ons but as integral to pedagogical design, particularly in disciplines that require open dialogue on sensitive topics.

Beyond pedagogical participation and theoretical contribution, the findings also highlight important design considerations for recording-mediated learning analytics systems. The increase in concern-related expression may indicate that visual privacy enabled participants to regulate identity boundaries while maintaining interactional presence, supporting socially engaged discourse practices emphasized in social constructivist accounts of collaborative learning. Students who might otherwise hesitate to contribute due to concerns about judgment or identity exposure may feel more comfortable articulating nuanced perspectives when their visual identity is abstracted. This consideration is particularly relevant as education increasingly adopts online and hybrid learning models, where video-based interaction amplifies visibility and self-consciousness.

Expression-preserving visual transformations can therefore function both as privacy-enhancing representations and as design strategies for ethically grounded learning analytics practices. Participants' reported comfort, perceived confidence, and willingness to engage in video capture suggest that reducing identity exposure while preserving expressive cues may help reconcile longstanding tensions between privacy and analytic utility in video-mediated learning environments. By operationalizing an expression-preserving facial transformation within an authentic collaborative context, this study extends prior technical work into a human-centered learning analytics setting, illustrating how privacy-transformed visual representations can support participation-sensitive analytics design.

Thus, this research contributes to the ongoing discourse on the ethical use of technology in education by illustrating how privacy-transformed visual representations can support participation while maintaining analytic relevance. The positive response to cartoonification suggests that offering learners greater control over their digital representation may serve as a meaningful pathway toward more ethically grounded and participation-sensitive educational technology.

5.4. Alternative explanations and design trade-offs

While the findings show that cartoonification supported expressive participation and increased willingness to engage in recorded discussions, several alternative mechanisms and design trade-offs should be considered when interpreting the results. These considerations do not negate the observed patterns but provide important context for understanding how visual privacy, discourse dynamics, and video-recording-mediated interaction intersect within learning environments.

Since participants in the cartoon-on condition experienced the visual transformation during the discussion, whereas the cartoon-off group did not, initial novelty or expectation effects may have contributed to engagement patterns. The introduction of a visually distinctive interface

can temporarily reduce inhibition or increase attentional engagement independent of sustained privacy perception. However, the persistence of concern-related expression across different topics and dyad-level analyses suggests that the observed shifts were not solely attributable to novelty. From a Privacy Regulation Theory perspective, novelty may serve as an initial catalyst for boundary recalibration, but the consistent distributional patterns suggest that perceived identity abstraction likely continued to shape expressive participation.

Visual privacy may reduce perceived identity exposure, but it can also reshape cues to interpersonal accountability in collaborative interaction. In social constructivist learning contexts, visibility and recognition often contribute to responsibility-taking and discourse regulation. Cartoonification may therefore recalibrate the balance between psychological safety and social accountability rather than simply increasing openness. The findings suggest that cartoonification support participation by lowering self-presentational pressure, yet they may also alter how learners negotiate authorship, evaluation, or peer perception during discussion. These sociotechnical trade-offs highlight the need to interpret increased expressive participation as a reconfiguration of interactional norms rather than a unidirectional improvement in engagement.

The emotionally charged nature of certain discussion prompts, particularly those involving social justice themes, represents another plausible explanatory factor. Although dyad-level analyses demonstrated consistent patterns across topics, some themes inherently elicited elevated emotional engagement regardless of visual condition. This suggests that expressive participation likely emerges from an interaction between visual privacy affordances and the ethical or affective demands of the discussion context. Rather than attributing outcomes solely to cartoonification, the results are best understood as reflecting how technological mediation interacts with the content of discourse to shape collaborative meaning-making processes.

Taken together, these alternative explanations underscore that cartoonification operates within a broader sociotechnical ecology involving technological affordances, participant expectations, and contextual dynamics. Acknowledging these mechanisms strengthens the interpretation of the findings by situating visual privacy not as a singular causal factor but as one component influencing how learners regulate identity boundaries and participate in socially constructed dialogue.

6. Limitations of the study

While this study provides important insights into the role of visual privacy in video-based multimodal learning analytics, several limitations should be considered when interpreting the findings. First, the sample comprised 36 participants from a single private university in the United States, which may limit generalizability across institutional contexts and learner populations. In addition, although statistical analyses were conducted at the dyad level to reduce the non-independence inherent in conversational interaction, dyads were nested within discussion topics and shared contextual conditions, which may introduce residual clustering effects. Given the modest number of dyads, multi-level or cluster-robust modeling was not applied; future work with larger samples could more explicitly model topic-level variation and interactional dependencies. Expanding the study to more diverse populations and instructional settings would strengthen the external validity of these findings and further clarify how visual privacy interventions function across varied collaborative environments. In addition, although three sensitive discussion topics were included (police brutality and racism, AI and racial bias, and tensions between science and religion), many other forms of sensitive dialogue remain unexplored. The effectiveness of cartoonification may vary depending on topic characteristics, cultural context, or discourse norms, and future work should examine a broader range of conversational settings.

Second, several sources of variability inherent to remote, discussion-based research may have influenced the observed outcomes. Although

GPU-accelerated facial detection reduced transformation failures, rare instances of undetected faces or skipped frames could occur under extreme head pose or lighting conditions. Emotional tone analysis relied on automated transcription and an emotion classification model trained on general-domain datasets; while applied consistently across conditions, such models may not fully capture the nuances of sensitive-topic dialogue or culturally specific forms of expression. Accordingly, emotion outputs in this study are interpreted as indicators of expressive participation within discourse rather than direct measures of internal emotional states. Furthermore, because discussions were intentionally unfacilitated to preserve natural conversational dynamics, dyads may have engaged with guiding questions at varying depths. These factors were mitigated through standardized prompts, consistent platform conditions, and uniform analytic procedures, but they remain considerations for future research using multimodal analytic pipelines.

Third, the study did not account for individual differences in baseline confidence, personality traits, or prior experiences with recorded discussion environments. Characteristics such as introversion, extroversion, or self-efficacy may shape expressive participation independent of privacy features. Future research could incorporate validated self-report instruments to better distinguish dispositional factors from the effects of visual privacy interventions. In addition, the findings capture participants' immediate responses to cartoonification, and longitudinal research is needed to examine whether the observed patterns persist over time or diminish as novelty effects fade. The study's focus on discussion-based activities also limits the scope of inference; future work should explore whether similar effects emerge in other collaborative learning contexts, such as presentations, design projects, or peer feedback sessions.

Fourth, although the cartoonification process substantially alters facial appearance while preserving expressive cues necessary for interaction, it should be interpreted as a privacy-enhancing visualization rather than a formally validated de-identification technique. Because certain facial structures are retained to support communicative expressiveness, the transformation may not prevent recognition under adversarial conditions. Our findings, therefore, speak to perceived visual privacy rather than to guaranteed anonymity. While preliminary emotion-detection comparisons suggest compatibility for affect-related tasks, future work should explore broader downstream analytic validation (such as gaze estimation or AU detection) and further evaluate the robustness of these visual transformations through human recognizability studies or algorithmic re-identification testing to better understand the boundary between identity abstraction and analytic utility.

Finally, this study's context was limited to discussion-based activities, and the effectiveness of cartoonification in other types of learning activities (e.g., presentations and group projects) remains to be explored. Additionally, this study primarily addressed visual privacy through cartoonification and did not examine other dimensions of privacy in digital learning environments, such as audio transformation, data storage governance, or multimodal anonymization strategies. Exploring alternative or customizable privacy-preserving representations may provide deeper insight into how learners negotiate participation, visibility, and trust within video-mediated collaborative settings.

7. Future work

The expression-preserving cartoonification approach introduced in this study opens several avenues for future research at the intersection of visual privacy and multimodal learning analytics. First, future work should examine how visual privacy influences analytic robustness across modalities by systematically comparing emotion-detection outcomes for cartoonized and non-cartoonized facial representations. Rather than treating cartoonification solely as a privacy intervention, such investigations can clarify how visual transformations reshape the signals available to multimodal models and how these transformations influence the interpretation of expressive participation. Preliminary analyses

using Py-Feat [66] suggest that emotion patterns derived from cartoonized faces align with text-based indicators, but more rigorous multimodal validation across datasets and models is needed.

Building on the sociotechnical trade-offs discussed in this work, future studies could also explore how visual privacy recalibrates interactional norms in learning environments. For example, integrating text, audio, and cartoonized visual features within multimodal fusion frameworks may help disentangle whether increased concern-related expression reflects reduced self-presentational pressure, shifts in accountability signaling, or broader changes in discourse dynamics. Also, longitudinal designs and controlled comparisons between different visual privacy transformations would further clarify how novelty, expectation effects, and evolving peer norms shape expressive participation over time.

Finally, cross-cultural and contextual investigations are essential for understanding how visual privacy is interpreted across diverse learning environments. Perceptions of identity exposure, anonymity, and social accountability vary across cultures, disciplines, and institutional settings. Comparative studies involving learners from different educational contexts could illuminate how privacy-enhancing visual representations are negotiated globally and inform the design of adaptable, participation-sensitive learning analytics systems that respect cultural expectations while preserving analytic value.

8. Conclusion

This study examined how expression-preserving cartoonification influences learners' willingness to participate in video-mediated collaboration and their expressive engagement during sensitive discussions. Motivated by the need to balance analytic value with student privacy concerns in multimodal learning analytics, the findings suggest that visual abstraction can support expressive participation by reducing perceived identity exposure while preserving interactional cues. Dyad-level analyses revealed a consistent shift toward concern-related expression when cartoonification was active, indicating a reconfiguration of how learners engaged with ethically complex topics rather than a generalized increase in emotional intensity. Complementary survey responses further suggested that perceived visual privacy was associated with greater willingness to participate in recorded activities, highlighting the role of design affordances in shaping participation conditions.

Rather than positioning cartoonification as a definitive privacy solution or a direct proxy for psychological constructs such as confidence, this work frames visual privacy as a sociotechnical mechanism that reshapes the balance between psychological safety and interpersonal accountability in collaborative dialogue. By operationalizing an expression-preserving facial transformation within an authentic learning analytics context, the study demonstrates how privacy-aware visual design can support participation-sensitive analytics without eliminating the communicative richness necessary for collaborative learning.

As video-based learning environments continue to expand, these findings underscore the importance of integrating privacy-enhancing representations into the design of pedagogical and analytical systems.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGpt and Claude AI tools to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and takes full responsibility for the publication's content.

CRediT authorship contribution statement

Tochukwu Dominic Eze: Conceptualization, Methodology, Software, Validation, Investigation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Khalil Anderson:** Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing. **Marcelo Worsley:** Writing – review & editing, Supervision, Project administration, Investigation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

No Funding was received for conducting this study

Financial Interest: The authors have no relevant financial or non-financial interests to disclose.

Appendix A

Fig. A.

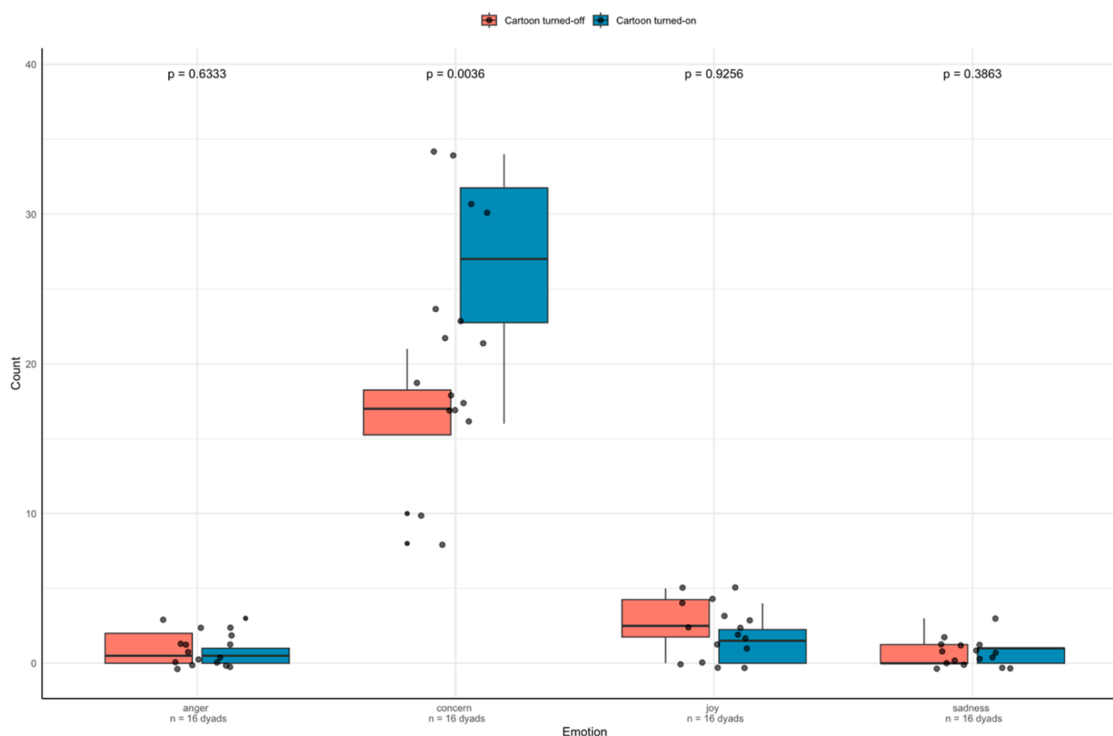


Fig. A. Emotion frequency distribution across the three topics, excluding the highest-value dyad.

As a robustness test, we conducted a sensitivity analysis excluding the dyad with the highest emotional expression value to determine whether the effect was disproportionately influenced by a single observation. Even after removal of this dyad, the difference between cartoon activation and deactivation remained statistically significant ($U = 6$, $p = 0.0036$, $r = 0.70$; $N_{on} = 8$, $N_{off} = 8$), with a large effect size. These results indicate that the observed difference reflects a stable pattern across dyads rather than being driven by an outlier.

Data availability

The datasets generated and/or analyzed during this study are not publicly available but can be obtained from the corresponding author upon reasonable request.

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